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DS 4002 - CS3
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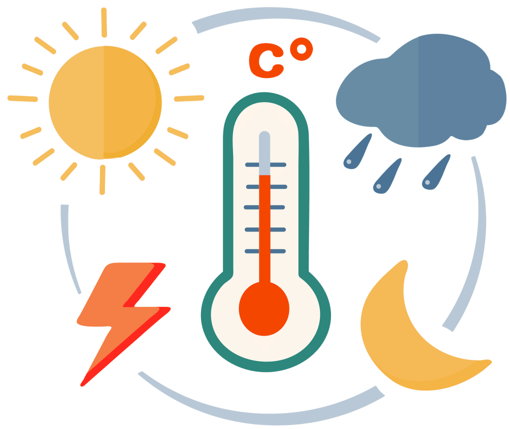
Predicting the 2023 Temperature Forecast in New York City



Heat-related deaths historically account for about 2% of all deaths occurring in the summer months in New York City. Due to climate change and increasing temperatures, this percentage is expected to go up if no preventative action is taken. The City of New York has asked you to project temperature forecasts an entire year in advance, so they are able to plan ahead for what they think may be the hottest NYC summer in recorded history, and create appropriate policies to counteract potential resulting heat-related deaths. You, as the data scientist on this 2022 case, are responsible for accurately forecasting the following year's daily temperature trends – your forecast will be handed to policymakers to decide what further action to pursue. You will be provided with daily maximum temperature measurements from 1970-2022 to use to build your model, which you should use to forecast the entire 2023 year. You will also be provided with corresponding precipitation measurements. The City of New York is curious about if precipitation significantly impacts temperature measurements, and if they should be conscious of this extraneous factor when preparing for extreme temperatures. You should also conduct an analysis to determine if precipitation is a useful predictor for temperature, to answer this question. A SARIMAX Time Series model may be useful for this task, as New York City experiences all four seasons with expected significant temperature fluctuations throughout the year.

The deliverable that is expected of you is a clear temperature forecast for the entire year of 2023. These daily temperature measurements should be aggregated to monthly averages to account for random variation that is non-seasonal. You may provide a confidence interval accompanying this exact forecast, to give policymakers an estimate that encompasses 95% of the variance they will see over the next year. This should be presented visually, in a way that is easy for non-data scientists to interpret. Additionally, the policymakers would like to know whether or not precipitation was a useful factor in creating this forecast. If it is not, this will save them some work and exploration later on, so it is an important factor to pay attention to. You should provide appropriate output to make this conclusion alongside your visualized forecast of 2023.

https://a816-dohbesp.nyc.gov/IndicatorPublic/data-features/heat-report-archive/2022/